THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 6: **Triangles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In $\triangle ABC$, $DE \parallel BC$ and $AD : DB = 1 : 2$. The ratio of areas of $\triangle ADE$ to trapezium $DBCE$ is 1: 1

(a) 4 **(b)** 10 **(c)** 8 **(d)** 9

2. In a right triangle with hypotenuse $c = 5$ and one leg $a = 3$, the area (sq. units) is: 1

(a) 6 **(b)** 12 **(c)** 10 **(d)** $\frac{15}{2}$

3. The altitude on the hypotenuse of the right triangle with legs 3 and 4 is of length (cm): 1

(a) 6 **(b)** $\frac{5}{2}$ **(c)** $\frac{7}{2}$ **(d)** $\frac{12}{5}$

4. If $\triangle ABC \sim \triangle PQR$, $ar(ABC) = 18$ and $ar(PQR) = 98$ and $AB = 6$, then $PQ =$ 1

(a) 98 **(b)** 14 **(c)** 6 **(d)** 17

5. In $\triangle ABC$ right-angled at B, $AB = 3$, $BC = 4$. If $BD \perp AC$, then AD (in cm) is: 1

(a) $\frac{16}{5}$ **(b)** $\frac{12}{5}$ **(c)** $\frac{9}{5}$ **(d)** $\frac{3}{2}$

6. A square has diagonal $5\sqrt{2}$ cm. Its area (cm²) is: 1

(a) 10 **(b)** 50 **(c)** 100 **(d)** 25

7. In $\triangle ABC$, $DE \parallel BC$ and $AD : DB = 2 : 3$. The ratio of areas of $\triangle ADE$ to trapezium $DBCE$ is 4: 1

(a) 29 **(b)** 21 **(c)** 25 **(d)** 9

8. In a right triangle with hypotenuse $c = 13$ and one leg $a = 5$, the area (sq. units) is: 1

(a) 60 **(b)** 30 **(c)** 78 **(d)** $\frac{65}{2}$

9. **Assertion (A):** In $\triangle ABC$, $DE \parallel BC$ with D on AB and E on AC. If $AD = 2$ cm, $DB = 3$ cm and $AE = 4$ cm, then $EC = 6$ cm. 1

Reason (R): A line parallel to one side of a triangle, meeting the other two sides at distinct points, divides those sides in the same ratio.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** In a triangle ABC right-angled at B, with $AB = 6$ cm and $BC = 8$ cm, the hypotenuse $AC = 10$ cm. 1

Reason (R): The Pythagoras theorem holds only for isosceles right-angled triangles.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. In $\triangle ABC$, $DE \parallel BC$, $AD = 3$ cm, $AB = 7$ cm and $BC = 14$ cm. Find DE. 2

12. The areas of two similar triangles are 27 cm² and 75 cm². If a side of the first is 3 cm, find the corresponding side of the second. 2

13. Find the altitude on the hypotenuse of the right triangle with legs 3 cm and 4 cm. 2

14. In $\triangle ABC$, $DE \parallel BC$, $AD = 2$ cm, $AB = 7$ cm and $BC = 21$ cm. Find DE. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In trapezium $ABCD$, $AB \parallel DC$ and the diagonals meet at O. If $AB = 8$ cm and $DC = 12$ cm, find $AO : OC$ and the ratio $ar(AOB) : ar(COD)$. 3

16. Find the altitude of an equilateral triangle of side 6 cm, in terms of $\sqrt{3}$. 3

17. $\triangle ABC$ is right-angled at B with $AB = 3$ cm and $BC = 4$ cm; $BD \perp AC$. Find AD, DC and BD. 3

OR

In trapezium $ABCD$, $AB \parallel DC$ and the diagonals meet at O. If $AB = 6$ cm and $DC = 10$ cm, find $AO : OC$ and the ratio $ar(AOB) : ar(COD)$. 3

18. Find the altitude of an equilateral triangle of side 8 cm, in terms of $\sqrt{3}$. 3

OR

$\triangle ABC$ is right-angled at B with $AB = 5$ cm and $BC = 12$ cm; $BD \perp AC$. Find AD, DC and BD . 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** At the same time of the day, a girl of height 1.5 m casts a shadow 2 m long, while a tower casts a shadow 24 m long on level ground. 5

(i) Name the similarity criterion that makes the two triangles formed (object, shadow, sun's ray) similar.

(ii) Find the height of the tower.

(iii) At that time, a pole 6 m high will cast a shadow of what length?

(iv) Find the distance from the top of the tower to the tip of its shadow.

20. Two ships leave a port at the same time; one sails 50 km due north and the other 120 km due west. Find the distance between them and the ratio of this distance to the one if both had sailed only half as far. 5

OR

Two poles of heights 45 m and 9 m stand on level ground 27 m apart. Find the distance between their tops. 5

END OF QUESTION PAPER — Total: 40 Marks